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PERFORMANCE SPECIFICATION
SCAN-INTERLACE, PL-1403/UA, PL-1403A/UA

This performance specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This performance specification covers one type of common module with two interchangeable models; Scan-Interlace (30Hz) PL-1403/UA and PL-1403A/UA.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-338/1	- Electronic Design Reliability
MIL-HDBK-454	- Standard General Requirements for Electronic Equipment.
MIL-HDBK-742	- Waste Disposal Methods for Military Packaging Materials
MIL-HDBK-781	- Reliability Testing for Engineering Development and Production Qualification

(Unless otherwise indicated, copies of the above specifications, standards and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094).

2.2.2 Other Government documents, drawings and publications. The following other government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2). The drawings listed below are to be used to ensure accurate interface of the item produced under this performance specification with existing hardware.

DRAWINGS

USA CECOM

SM-D-657596	- ESD, Scan-Interlace PL-1403A/UA
SM-D-657597	- PWB, Scan-Interlace PL-1403A/UA
SM-D-657598	- Scan-Interlace PL-1403A/UA
SM-D-657599	- Scan-Interlace PL-1403A/UA Outline Dimensions and Installation
SM-C-772134	- Jack, Tip
SM-C-772278	- Connector, Receptacle, Electrical
SM-C-773501	- Connector, Receptacle, Electrical
SM-D-773948	- Schematic Diagram, Scan and Interlace
SM-D-773949	- Printed Wiring Board, Scan and Interlace 30 Hz
SM-D-773950	- Scan-Interlace (30Hz) PL-1403/UA

(Copies of documents, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 122-89 - Sample Size to Estimate a Measure of Quality for a Lot or Process

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/IPC S-815A - General Requirements for Soldering Electronic Inter-Connections

(Application for copies should be addressed to the American National Standards Institute Inc., 1430 Broadway, N.Y., N.Y. 10018).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ANSI/ASQC Z 1.4 - Sampling Procedures; Tables for Inspection by Attributes

(Application for copies should be addressed to the American society for Quality Control, Inc. P.O. box 3005, 611 East Wisconsin Avenue, Milwaukee, WI, 53202).

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Where specified in the contract (see 6.2), the contractor shall furnish first article units for inspection (see 4.2).

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3.2 Materials. The contractor shall select the materials, but the materials and parts used shall be fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria of the class, grade or type part shall be that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The Scan-Interlace PL-1403/UA and PL-1403A/UA, consists of a printed wiring board (PWB) and circuitry to drive a mechanical scanner at 30Hz scan rate in low power infrared systems. The two models differ in their internal configuration but are identical in performance and external configuration. In system usage, they are, and must be, completely interchangeable.

3.3.1 Design. The Scan-Interlace shall meet form, fit, function and interface requirements as specified herein.

3.3.2 Weight. The weight of the Scan-Interlace shall be equal to or less than 0.26 pound (4.16 ounces) when weighed with a balance or scale having an accuracy of 1% or better.

3.3.3 Dimensions. The Scan-Interlace Printed Wiring Board (PWB) shall be of appropriate material with a nominal thickness of 0.062 inches and shall have dimensions of 3.02 inches x 3.48 inches, $\pm$ 0.02 inches. When mounted, components shall not extend more than a maximum of 0.414 inches from the surface of the PWB. Two corners of the board on the 3.02 inch side away from the two large connectors shall be trimmed at a 45° angle between two points 0.25 inches vertical and horizontal of the corners, removing a triangle of board material 0.25 by 0.25 on either side of the right angle (corner). The x-axis datum line shall be located parallel to the 3.48 inch edge of the board and 0.125 inches above that edge, with the two large connectors above it. It shall pass through the two 0.094 inch diameter holes located 0.125 inches above the edge and 0.290 inches and 3.190 inches in from the square edge. The y-axis datum line shall be located 0.290 inches in from the left-hand edge of the board. It shall pass through the two left-hand 0.094 inch diameter holes located 0.290 inches in from the left edge and shall be perpendicular to the x-axis datum line. Four mounting holes 0.140 inches in diameter shall be located 0.025 inches and 2.750 inches above the x-axis datum line and 0.245 inches and 2.370 inches to the right of the y-axis datum line as shown on Drawing SM-D-773949 or SM-D-657597. Two connectors shall be attached to the non-component side of the board for the purpose of interfacing with existing hardware. These connectors shall have the dimensions of Part Numbers SM-C-772278-3 and -4 as shown on Drawing SM-C-772278, while the third connector, which is on the component side of the board, shall have the dimensions of Part Number SM-C-773501-2 as shown on Drawing SM-C-773501. These connectors shall be located approximately as shown on Drawing SM-D-773950 or SM-D-657598 for P1, P2 and J1. Two tip jacks shall also be attached to the component side of the board. These tip jacks shall mount approximately where shown on Drawing SM-D-773950 or SM-D-657598. The tip jacks shall

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meet the physical and electrical parameters of part numbers SM-C-772134-1 and -2 in accordance with Drawing SM-C-772134 for the PL-1403/UA and part numbers SM-C-772134-2 and -6 in accordance with Drawing SM-C-772134 for the PL-1403A/UA. They shall be located as shown on SM-D-773950 or Drawing SM-D-657598. The pinout assignments for the connectors shall be in accordance with Drawing SM-D-773948 or Drawing SM-D-657596 to ensure proper interface with existing hardware.

3.4 Components. The Scan-Interlace shall contain manufactured components selected by the contractor as required to meet the performance specifications contained herein.

3.5 Performance Characteristics.

3.5.1 *Power consumption. The Scan-Interlace shall operate with the following voltages (see Figure 1), and total power shall not be greater than 2.2 watts:

a. + 4.8, ±0.1 Vdc

b. - 4.8, ±0.1 Vdc

c. +10.0, ±0.1 Vdc and -10.0, ±0.1 Vdc

* Power shall be measured as the sum of the products of the power supply voltage and their average currents.

3.5.2 Identification resistor verification. The Scan-Interlace shall contain two identification resistors as shown on SM-D-773948 or SM-D-657596.

3.5.3 Regulator voltage. The Scan-interlace shall contain a positive voltage regulator adjustable to +3.40, ±0.05 Vdc and a negative voltage regulator adjustable to -3.40, ±0.05 Vdc.

3.5.4 Transducer bridge voltage. The Scan-Interlace shall provide a transducer bridge voltage which is adjustable to +1.25, ±0.05 Vdc (NOTE: When this voltage is indicated, the actual bridge voltage, i.e., voltage applied across the bridge, is +2.15 Vdc). An average voltage shall be measured at this test point as per 4.5.4.

3.5.5 Scan motor drive circuitry. The Scan-Interlace shall provide for adjustment of the scan drive waveform to an amplitude of 1.5, ±0.5 volts peak-to-peak (P-P) during the active portion (see Figure 2).

3.5.6 Interlace position amplifier. The Interlace Position Amplifier shall provide a gain of 100, ±10 volts per volt for signals generated by the Scanner Interlace Position Transducer.

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3.5.7 Position reference circuitry. The Scan-Interlace shall provide circuitry for generating a position reference waveform adjustable to an amplitude of 0.6, ±0.2 volts P-P (see Figure 3).

3.5.8 Position error circuitry. The Scan-Interlace shall provide circuitry for generating a position error waveform adjustable to 0.40, ±0.15 volts P-P (see Figure 4).

3.5.9 Rate reference circuitry. The Scan-Interlace shall provide circuitry that will produce a rate reference waveform adjustable to an amplitude of 0.6, ±0.2 volts P-P (see Figure 5).

3.5.10 Solenoid drive circuitry. The Scan-Interlace shall provide circuitry to develop and to equalize the scanner solenoid drive waveforms, with an amplitude of 0.10, ±0.05 volts (see Figure 6).

3.5.11 Video gate. The Scan-Interface shall provide video gate electronics capable of generating a video gate pulse width adjustable to 2.0, ±0.2 milliseconds.

3.5.12 Side/rear phasing. The Scan-Interlace shall accept and process the tachometer and interlace position signals in both side and rear phase relationships when properly interconnected (see Figures 6, 7, 8, 9 and 10).

3.6 Environmental conditions.

3.6.1 Temperature shock. The Scan-Interlace shall not be damaged (see 6.4.1) by sudden changes in temperature between -54°C and +95°C.

3.6.2 High temperature. The Scan-Interlace shall not be damaged by storage to +95°C or continuous operation to +71°C and shall operate for 30 minutes at +95°C. The Scan Drive waveform amplitude (Figure 2) shall not change more than ±40% from the initial setup at ambient temperature over the specified operating temperature range (+71°C). The Solenoid Drive waveform amplitude (Figure 6) shall not change more than ±25% from the initial setup at ambient temperature over the specified operating temperature range (+71°C).

3.6.3 Low temperature. The Scan-Interlace shall not be damaged by operation to -54°C, or storage to -62°C. The Scan Motor Drive waveform amplitude (Figure 2) shall not change more than ±40% from the initial setup at ambient temperature over the specified operating temperature range (-54°C). The Solenoid Drive waveform amplitude (Figure 6) shall not change more than ±25% from the initial setup at ambient temperature over the specified operating temperature range (-54°C).

3.6.4 Shock. The Scan-Interlace shall not be damaged by high intensity shock of 100 g's peak amplitude with 11 milliseconds duration and by bench handling test.

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3.6.5 Vibration. The Scan-Interlace shall not be damaged by vibration over the frequency spectrum at the specified "g" levels and amplitudes shown in Figure 11.

3.7 Reliability. The Scan-Interlace shall have a mean time between failure (MTBF) of at least 17,500 hours (θ_0). Use of the procedures described in MIL-HDBK-338/1 and evaluation methods per MIL-HDBK-781 are recommended (see 4.7).

3.8 Coating. Unless otherwise specified, the assembly and its components and parts, shall be cleaned and shall be protected by a conformal insulating coating.

3.9 Preconditioning. Each Scan-Interlace shall be subjected to at least a 20-cycle preconditioning environment as specified in Figure 12 prior to group A inspection. For an exception to this requirement, see 6.2.

3.10 Soldering. Unless otherwise specified, electronic soldering shall conform to ANSI/IPC-S-815A.

3.11 Nameplates and product marking. The assembly, parts, components, subassemblies and assemblies, thereof, shall be marked for identification and reference designation. Requirement 67 of MIL-HDBK-454 may be used for guidance.

3.12 Workmanship. All electronic parts, components, assemblies or sub-assemblies shall be free of smudges, excess solder, metal chips or the existence of any foreign material on any surface. Bearing assemblies shall be free from rust, dirt or tool marks. Wires and integrated circuitry shall be protected from contact with rough or irregular surfaces and be shielded from shorting.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.4).

4.2 First article inspection. First article inspection shall be performed by the contractor on a specified number of systems when a first article inspection is required (see 3.1). First article reliability testing per 4.7 will not be required unless specified in the contract or purchase order.

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4.2.1 Inspections. All materials, parts, processes and assemblies shall be visually examined for conformance to the applicable specification or drawing. Inspections shall be made using ANSI/ASQC Z 1.4 sampling procedures. Sample size shall conform to ASTM E 122-89 as cited in the contract or purchase order (see 6.2).

TABLE I. Visual Inspections.

Inspection Requirement	Requirement paragraph	Inspection criteria
Any part or component missing or damaged	3.3	4.2.1
Weight not as specified	3.3.2	4.2.1
Coating not as specified	3.8	4.2.1
Soldering not as specified	3.10	4.2.1
Marking not as specified	3.11	4.2.1
Workmanship not as specified	3.12	4.2.1
Dimensions not as specified Length Width Height (maximum) Solder/lead protrusion Pins (connector location) Pins (guide location)	3.3	4.2.1

4.2.2 Tests. Upon successful completion of inspections specified in 4.2.1, perform the tests of Table II in any order. Failure of any tests shall constitute first article failure.

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TABLE II. First article inspection.

Inspection paragraph	Requirement paragraph	Test paragraph
Power consumption	3.5.1	4.5.1
Identification resistor verification	3.5.2	4.5.2
Regulator voltage	3.5.3	4.5.3
Transducer bridge voltage	3.5.4	4.5.4
Scan motor drive circuitry	3.5.5	4.5.5
Interlace position amplifier	3.5.6	4.5.6
Position reference circuitry	3.5.7	4.5.7
Position error circuitry	3.5.8	4.5.8
Rate reference circuitry	3.5.9	4.5.9
Solenoid drive circuitry	3.5.10	4.5.10
Video gate	3.5.11	4.5.11
Side/rear phasing	3.5.12	4.5.12
Temperature shock	3.6.1	4.6.1
High temperature	3.6.2	4.6.2
Low temperature	3.6.3	4.6.3
Shock	3.6.4	4.6.4
Vibration	3.6.5	4.6.5
Reliability	3.7	4.7

4.2.3 Disposition of first article samples. First article samples shall not be considered as part of the procurement quantities (see 6.2).

4.3 Inspection conditions. Unless otherwise specified herein, the assembly shall meet the specified performance requirements at ambient temperature (see 6.4.2).

4.3.1 Test power supplies. The test power supplies shall provide the voltages as specified in 3.5.1.

4.3.2 Supply ripple. The supply ripple shall not be greater than 0.1 volts P-P for the +4.8 Vdc supplies, and shall not be greater than 0.2 volt P-P for the +/-100.0 Vdc supplies.

4.3.3 Supply rise time. The supply rise times shall not be greater than 0.5 seconds.

4.3.4 Supply tracking. Turn-on rise times of the ±10.0 Vdc power supplies shall not be greater than that of the ±4.8 Vdc power supplies. The rise times of the ±4.8 Vdc power supplies shall track within 20 percent.

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4.3.5 Impedance. The supply output impedances shall not be greater than 0.1 ohm to 10 kHz and 5 ohms to 100 kHz.

4.4 Conformance inspection.

4.4.1 Visual inspection. Inspection shall be the same as specified in 4.2.1.

4.4.2 Tests.

4.4.2.1 Group A inspection. Group A inspection shall be conducted at an ambient temperature of (see 6.4.2) on all Scan-Interlaces following the preconditioning of 4.8 and the inspection 4.4.1. Group A tests may be performed in any order. Failure of any test shall be cause for rejection of the unit.

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TABLE III. Group A inspection.

			Scan-Interlace PL-1403/UA or PL-1403A/UA	
Inspection	Reqt para.	Test para.	Adjustment during test	Pins tested
Power consumption	3.5.1	4.5.1	None	None
Identification resistors	3.5.2	4.5.2	None	P1-4, P1-16 P1-6, P1-16
± 3.4 Vdc regulator voltage	3.5.3	4.5.3	+REG R6 -REG R7	+J1-8 P2-14 +J1-11 P2-14
Transducer bridge voltage	3.5.4	4.5.4	XDCR BR R14	J1-17(+) PL-1403/UA J2(+) PL-1403A/UA P2-14(-)
Scan motor drive waveforms	3.5.5	4.5.5	Scan speed R24	J1-18 J1-19
Interlace position amplifier waveform	3.5.6	4.5.6	None	J1-1 or J3
Position reference waveform	3.5.7	4.5.7	Pos. error gain R46	J1-2
Position error waveform	3.5.8	4.5.8	Pos. error gain R42	J1-5
Rate reference waveform	3.5.9	4.5.9	Pos. error gain R65	J1-12 PL-1403/UA J1-9PL-1403A/UA
Solenoid drive waveform	3.5.10	4.5.10	Bal R64	J1-3 J1-4
Video gate width	3.5.11	4.5.11	Gate width R57	J1-14 PL-1403/UA J1-7 PL-1403A/UA
Side/rear phasing	3.5.12	4.5.12	None	J1-3 J1-13 J1-14

4.4.2.2 Group B inspection. Not required.

4.4.2.3 Group C inspection. Group C inspections shall be conducted on assemblies selected from units which have passed the tests in 4.4.2.1. The sample (s) shall be subjected to the inspection listed in Table IV. Samples shall be selected in accordance with the contract or purchase order.

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4.4.2.3.1 Sampling for Group C inspection. Group C inspection samples shall be selected as specified in the contract or purchase order.

TABLE IV. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Vibration	3.6.5	4.6.5

4.4.2.3.2 Group C failures. Actions required relative to group C failures shall be as specified in the contract (see 6.2).

4.4.2.3.3 Disposition of Group C samples. Group C samples shall be accepted on contract subsequent to the tests of table III.

4.4.2.4 Group D inspection. Not required.

4.5 Methods of inspection. The waveforms and voltages necessary to test performance characteristics are shown in Figures 1, 7, 8, 10, and 13. The following performance characteristics can be measured after these waveform and voltage preparations have been completed.

4.5.1 Power consumption. The Scan Interlace shall be set up according to 3.5.1 through 3.5.12. The current being drawn by the assembly from each voltage source shall be measured with a device that will obtain the average current being drawn. The power delivered shall be calculated as the sum of the products of the power supply voltages and their average currents, as found above. Failure to meet the requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Identification resistor verification. Using a proper measuring device, measure the resistance between P1-4 and P1-16 for Identification Resistor number 1, and between P1-6 and P1-16 for Identification Resistor number 2. Failure to meet the requirements of 3.5.2 shall constitute failure of this test.

4.5.3 Regulator voltage. Using a proper measuring device, measure the +3.4 Vdc regulator voltage at J1-8 and the -3.4 Vdc Regulator Voltage at J1-11 using P2-14 as reference ground. Failure to meet the requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Transducer bridge voltage. Monitor the voltage between J1-17(+) (PL-1403/UA) or J2 (PL-1403A/UA) and P2-14(-) using a proper device. An average voltage shall be measured at this test. Failure to meet the requirements of 3.5.4 shall constitute failure of this test.

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4.5.5 Scan motor drive circuitry. Monitor the scan drive waveforms at J1-18 and J1-19 using an oscilloscope. Measure the peak-to-peak amplitude of the active portions of the scan drive waveforms as indicated in Figure 2. Failure to meet the requirements of 3.5.5 shall constitute failure of this test.

4.5.6 Interlace position amplifier gain. Measure the interlace position waveform at the input of the assembly, P1-14 (Hi) and P1-13 (Lo) with an oscilloscope. Also measure the interlace position waveform at the output of amplifier AR4 (J101). The gain is computed by the following formula:

gain = voltage at J1-1 (P-P) /input voltage (P-P).

Failure to meet the requirements of 3.5.6 shall constitute failure of this test.

4.5.7 Position reference. Monitor the position reference waveform at J1-2 with an oscilloscope. Measure the peak-to-peak amplitude of the waveform. To measure, first monitor the position reference waveform at J1-2 with an oscilloscope, then measure the peak-to-peak amplitude of the waveform. Failure to meet the requirements of 3.5.7 shall constitute failure of this test.

4.5.8 Position error. Monitor the position error waveform at J1-5 with an oscilloscope. Measure the peak-to-peak amplitude of the waveform. Failure to meet the requirements of 3.5.8 shall constitute failure of this test.

4.5.9 Rate reference. Monitor the rate reference waveform at J1-12 (PL-1403/UA) or J1-9 (PL-1403A/UA) with an oscilloscope. Measure the peak-to-peak amplitude of the waveform. Failure to meet the requirements of 3.5.9 shall constitute failure of this test.

4.5.10 Solenoid drive circuitry. Monitor the solenoid current waveforms at J1-3 and J1-4 using an oscilloscope. Measure the waveform amplitudes and compare to the requirements of 3.5.10. Failure to meet the requirements of 3.5.10 shall constitute failure of this test.

4.5.11 Video gate. Monitor the video gate at J1-14 (PL-1403/UA) or J1-7 (PL-1403A/UA) using an oscilloscope. Measure the gate width and compare to the requirements of 3.5.11. Failure to meet the requirements of 3.5.11 shall constitute failure of this test.

4.5.12 Side/rear phasing. Ensure that the input tachometer and interlace position signals are properly phased (see Figure 9) and that the side/rear switch (see Figure 10) is in the side position. Then, monitoring with an oscilloscope, compare the waveform at J1-13 to J1-4. Repeat measurement with the input signals and switch setting for rear phasing. The relationship shall be as described in 3.5.12. Failure to meet the requirements of 3.5.12 shall constitute failure of this test.

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4.6 Environmental tests. Environmental testing shall be performed in accordance with the requirements of this specification, except when a contractor's own test plan showing equivalent methods has been approved by the contracting office. When the contractor's test plan deviates from the test procedures prescribed, the contractor shall be required to show where his own tests test methods for verifying the specified performance are equivalent. Also, measurements taken by the contractor during verification testing of all environmental requirements shall be accomplished using known testing apparatus such as an environmental test chamber, vibration test table, pendulum shock device or similar test equipment for which the technical capabilities to properly simulate extreme climatic or transportation conditions are scientifically documented. The methods for testing to verify compliance with all stated performance characteristics selected by the contractor shall be equivalent to the recommended procedures found herein.

4.6.1 Temperature shock. Measure compliance with the requirement of 3.6.1 via performing the tests prescribed in 6.3. Failure to meet the requirement shall constitute failure of the sample.

4.6.2 High temperature. Measure compliance with the requirement of 3.6.2 via performing the tests prescribed in 6.3. Failure to meet the requirement shall constitute failure of the sample.

4.6.3 Low temperature. Measure compliance with the requirement of 3.6.3 via performing the tests prescribed in 6.3. Failure to meet the requirement shall constitute failure of the sample.

4.6.4 Shock. Measure compliance with the requirement of 3.6.4 via performing the tests prescribed in 6.3. Failure to meet the requirement shall constitute failure of the sample.

4.6.5 Vibration. Measure compliance with the requirement of 3.6.5 via performing the tests prescribed in 6.3. Failure to meet the requirement shall constitute failure of the sample.

4.7 Reliability.

4.7.1 First article reliability. The lower MTBF (θ_0) shall be demonstrated in accordance with the following sub-paragraphs:

4.7.1.1 Test Conditions.

a. Temperature: +50°C, (+5 -0)°C.

b. Temperature cycling: None.

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c. Vibration: 2.2 g, ±10 percent peak acceleration value at any non-resonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. The vibration shall be normal to the plane of the printed circuit board.

d. Equipment on-off cycling: Turn equipment on and let the temperature stabilize. Hold for 3 hours, then turn the equipment off and let the temperature stabilize. This cycle shall continue throughout the test.

4.7.1.2 Parameters. The discrimination ratio θ_0 / θ_1 shall be 3. For guidance, use MIL-HDBK-781.

4.7.1.3 Failure criteria and test length. The tests of Table III shall be performed at least once prior to and once after the tests of 4.4.2.3.1. Additional testing of Table III may be selected by the contractor. The results of all Table III tests and any physical damage shall be used to determine the success or failure of the reliability test. In the event of a Table III test failure, the time interval between the preceding successful test and the failure shall not contribute to the total cumulative testing hours. Testing shall continue until the total unit hours, together with the total count of relevant equipment failures, permit either an accept or reject decision in accordance with Figure 14. Only equipment "on" time shall be less than one-half the average operating time of all equipment on test.

4.8 Preconditioning. Each Scan-Interlace, in a non-operating mode, shall be subjected to 20 cycles as specified in Figure 12. Preconditioning may be performed before or after conformal coating.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's Packaging Activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. The Scan-Interlace, PL-1403/UA and PL-1403A/UA is intended for use in infrared systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. First article tests required. Also include:

1) Time frame for submission of the first article test reports when first article testing is required.

2) Time frame for the approval of first article test reports when first article tests are required.

3) First article reliability testing, if required.

4) Disposition of samples, when required.

c. Production delivery schedule.

1) Defined in terms of monthly lots.

2) Actions required relative to Group C failures (see 4.4.2.3.2).

d. Level A or level B preservation and packaging (see section 5).

e. Methods approved for use during environmental testing (see 4.6).

f. The preconditioning requirement 3.9 and inspection are required for procurement of all equipment (s) to this specification made directly for the Government. Implementation of 3.9 may be waived by the system contractors procuring equipment(s) to this specification for use in the manufacture of complete systems if the systems or subassemblies thereof are subjected to equivalent preconditioning.

g. Required number of test samples for first article testing and conformance testing.

h. Environmental pollution prevention measures as contained in the packaging material specifications referenced herein. Refer to MIL-HDBK-742 for disposal methods for packaging materials.

i. When first article rough handling tests are not required.

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j. Issue of DoDISS, to be cited in the solicitation, and if required, the specified issue of individual documents referenced (see 2.2.1 and 2.2.2).

6.3 Environmental inspection procedures. The test methods and procedures of MIL-STD-810 listed below are recommended, but not mandatory for determining compliance with the environmental requirements of this specification. The contractor may use any test method that he can verify as providing comparable stress levels on the material to be tested.

Inspection	Spec. Paragraph	MIL-STD-810 Procedure	MIL-STD-810 Procedure	Remarks
Temperature shock	3.6.1	503.1	I	Steps 1 & 4 shall be +95°C and Step 2 shall be -54°C transition from +95°C to -54°C and from -54°C to +95°C shall include a 2 hour stabilizing period at +23°C.
High temperature (storage)	3.6.2	501.1	I	Storage temperature (Step 2) shall be +95°C for 6 hours duration.
High temperature (operation)	3.6.2	501.1	I	Continuous operation (Step 4) shall be +71°C.
Low temperature (storage)	3.6.3	502	I	Storage temperature of Step 2 shall be -62°C for 9 hours.
Low temperature (operation)	3.6.3	502	I	Operation temperature is -54°C.
Shock Vibration	3.6.4 3.6.5	516.2 514.2	II VIII	Time duration: 11 milliseconds. Use Curve V. As an alternative, use method 201A of MIL-STD-202.

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6.4 Definitions.

6.4.1 Damage. Breakage, loosening, shifting, evidence of corrosion or failure of any finish, hardware, connection or component; any degradation of assembly performance to values less than specified.

6.4.2 Ambient temperature. Ambient temperature is 23°C ±5°C.

6.5 Subject term (key word) listing.

Infrared

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CUSTODIAN:
Army-CR

PREPARING ACTIVITY:
Army-CR

(Project 5855-0121)

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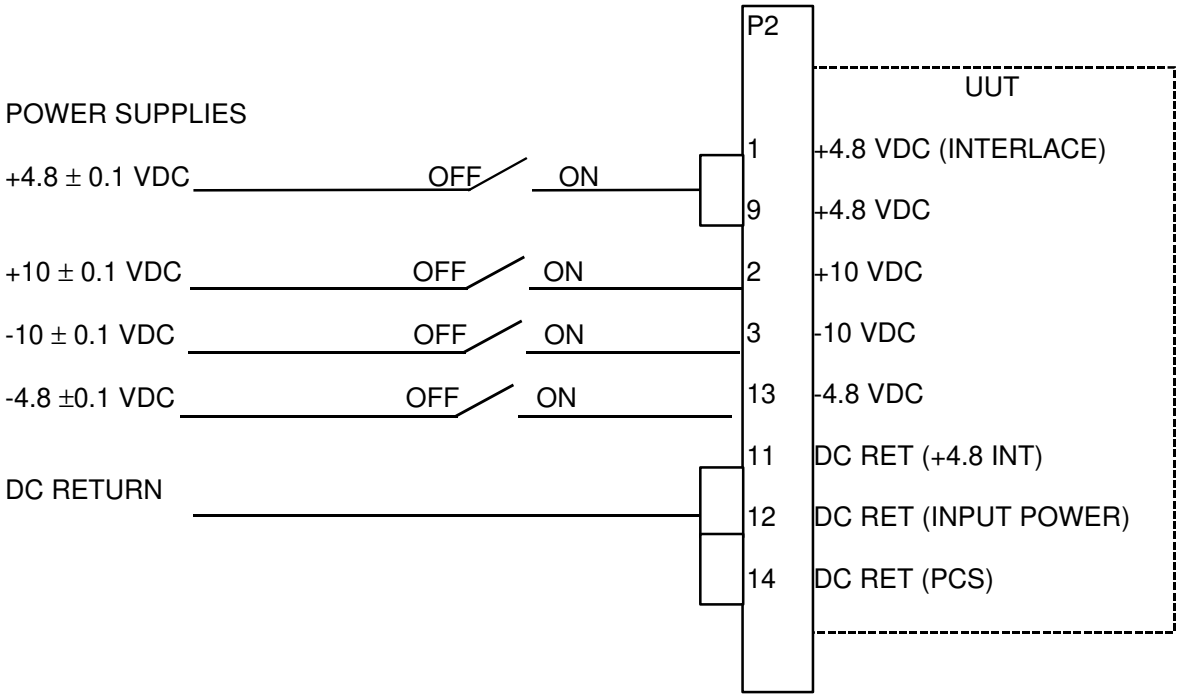


FIGURE 1. Input Voltage Preparation.

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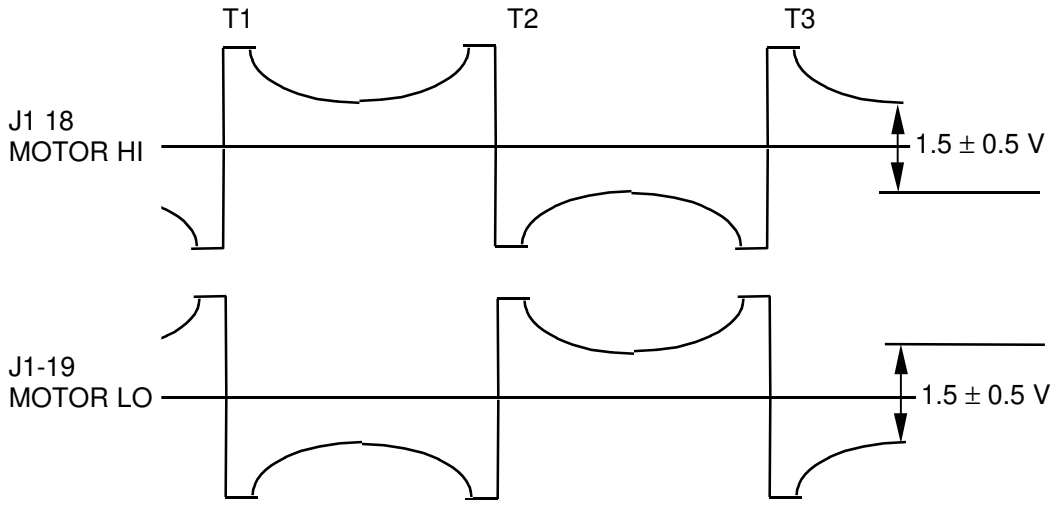


FIGURE 2. Scan Motor Drive Wave Forms

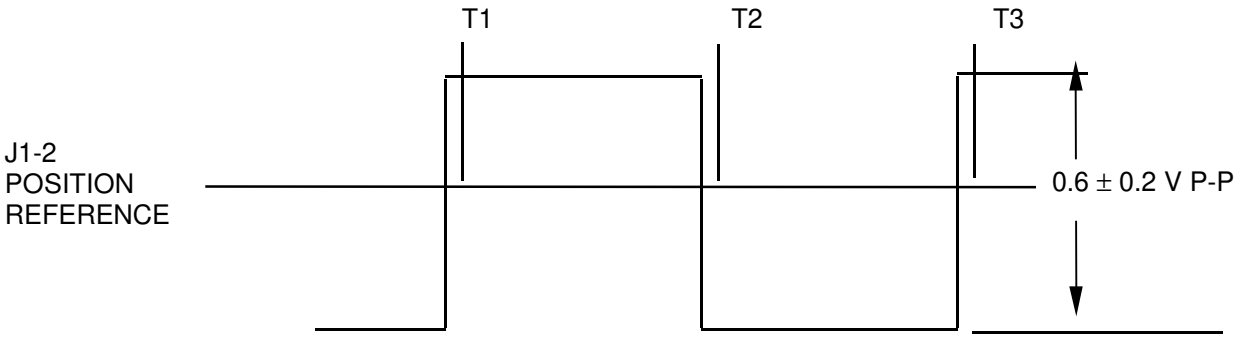


FIGURE 3. Position Reference Waveform.

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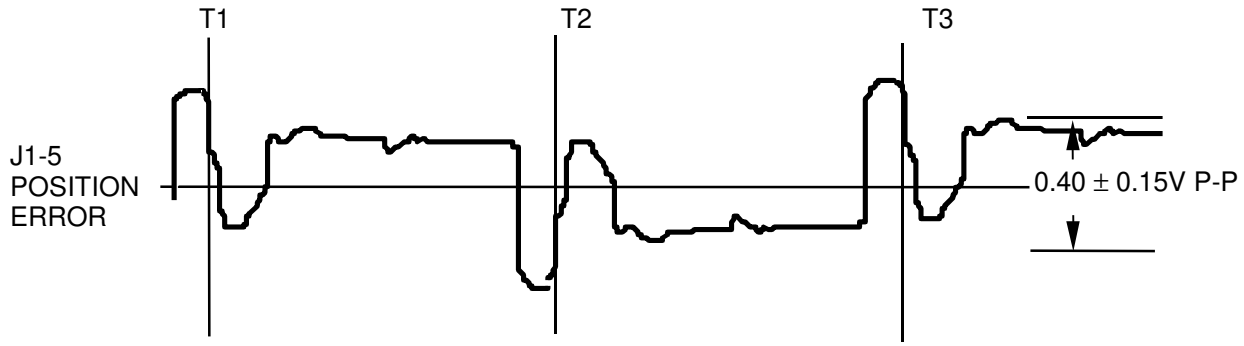


FIGURE 4. Position Error Waveform.

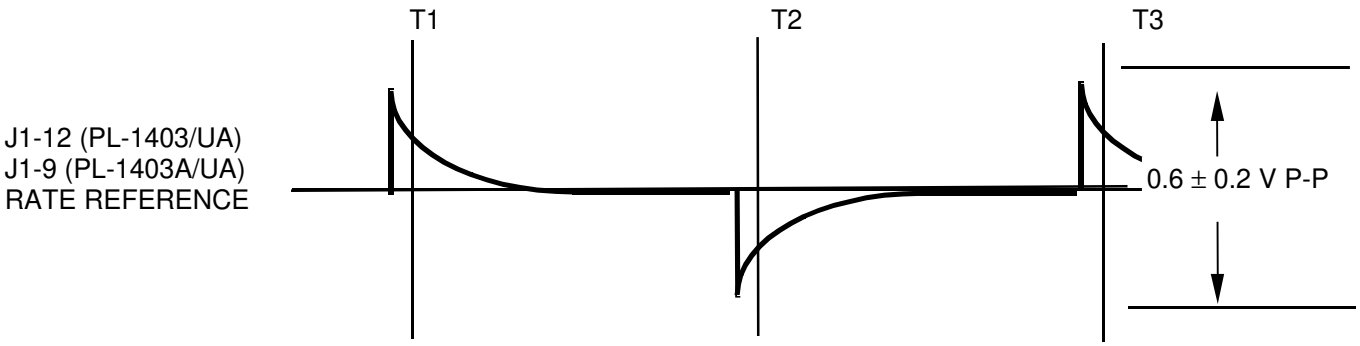


FIGURE 5. Rate Reference Waveform

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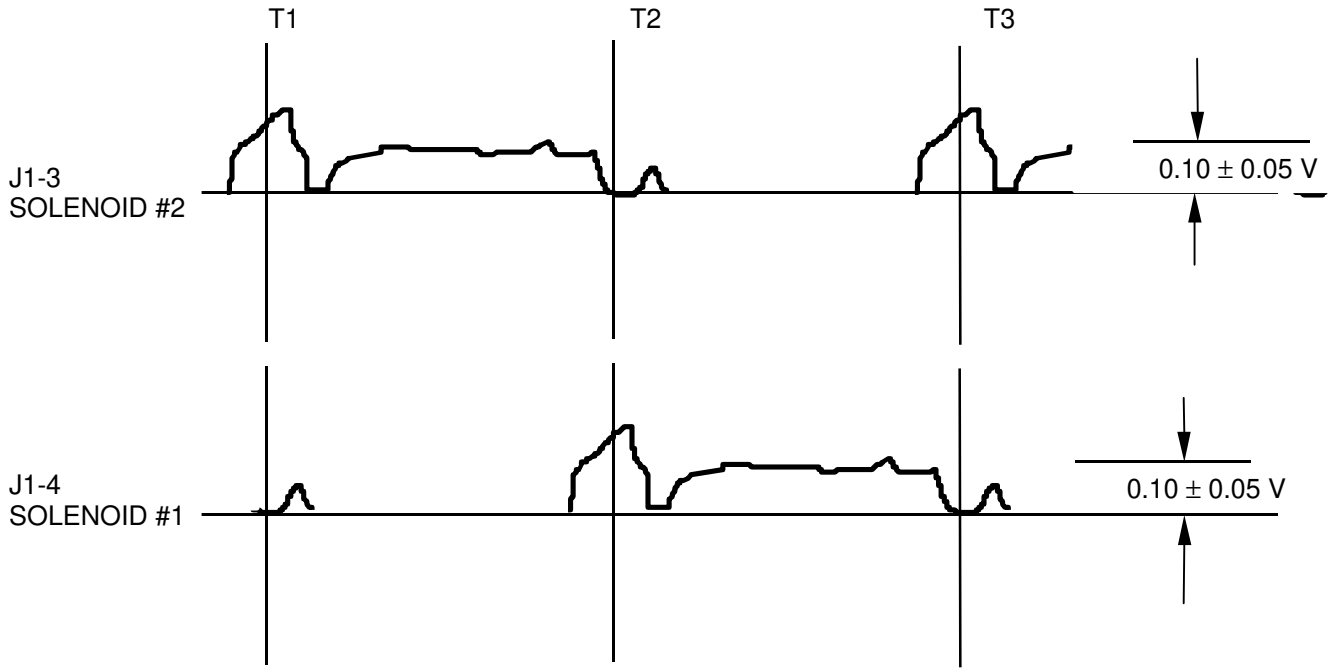


FIGURE 6. Solenoid Current Waveforms (Rear Phasing).

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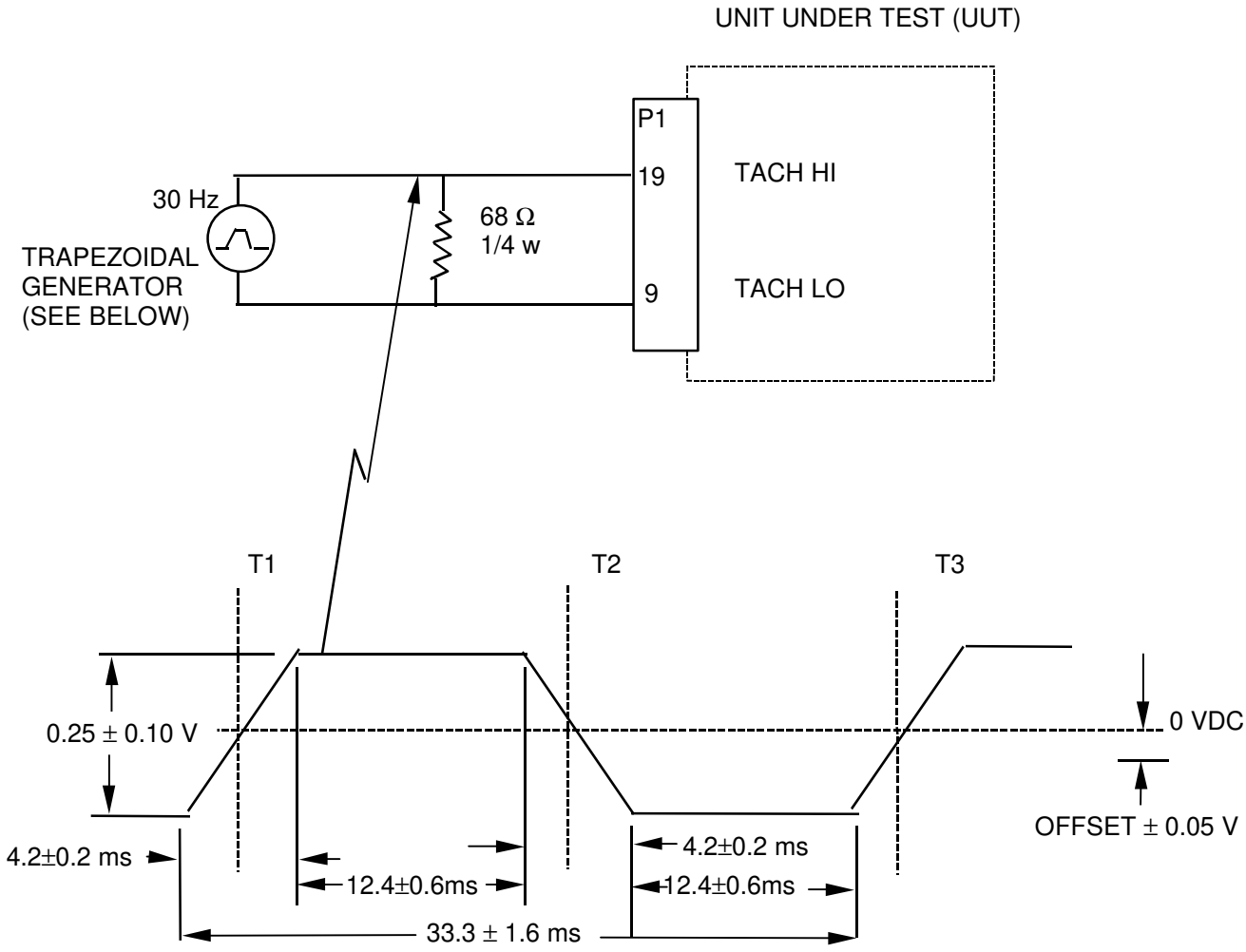


FIGURE 7. Tachometer Input Signal (Rear Phasing).

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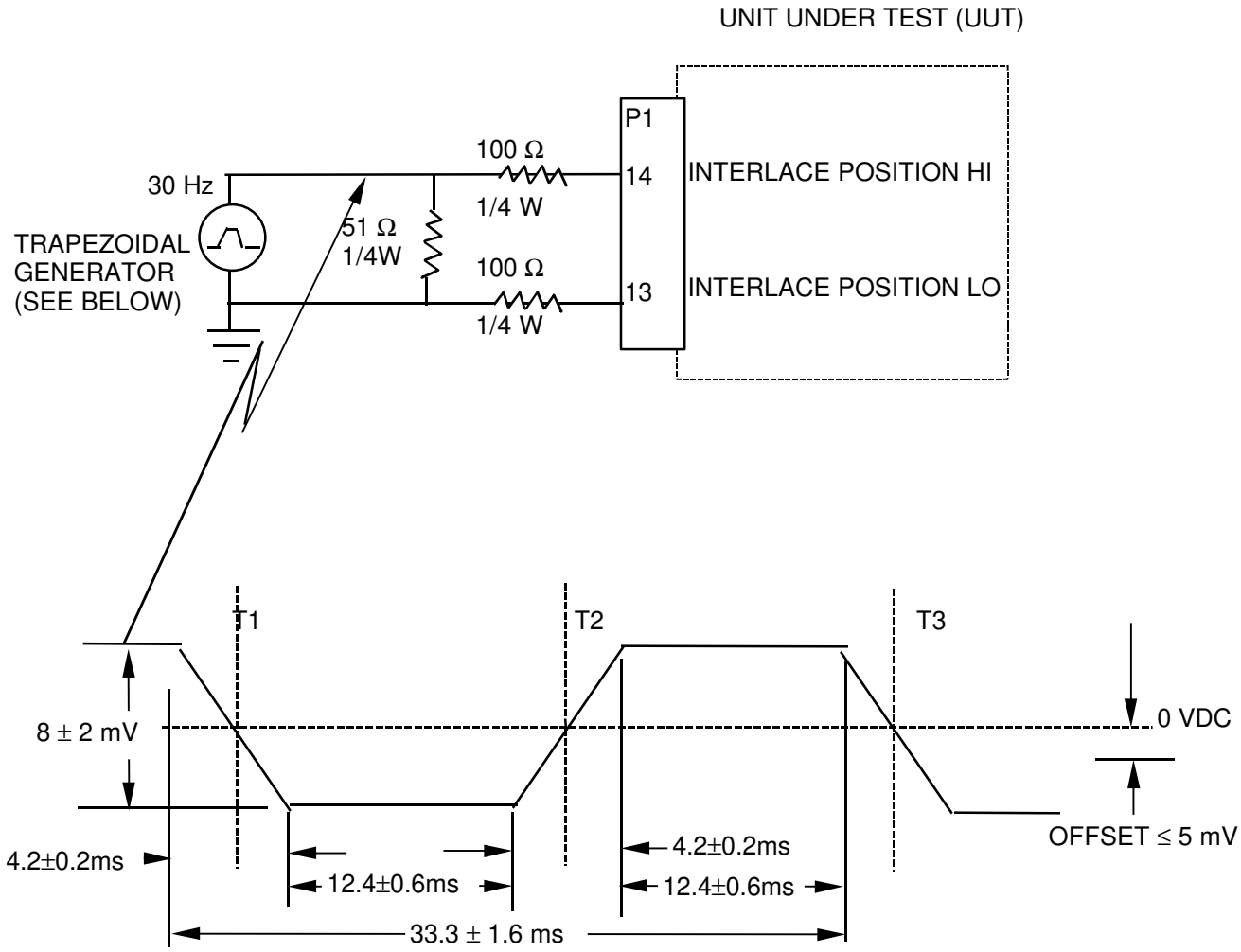


FIGURE 8. Interlace Position Input Signal (Rear Phasing).

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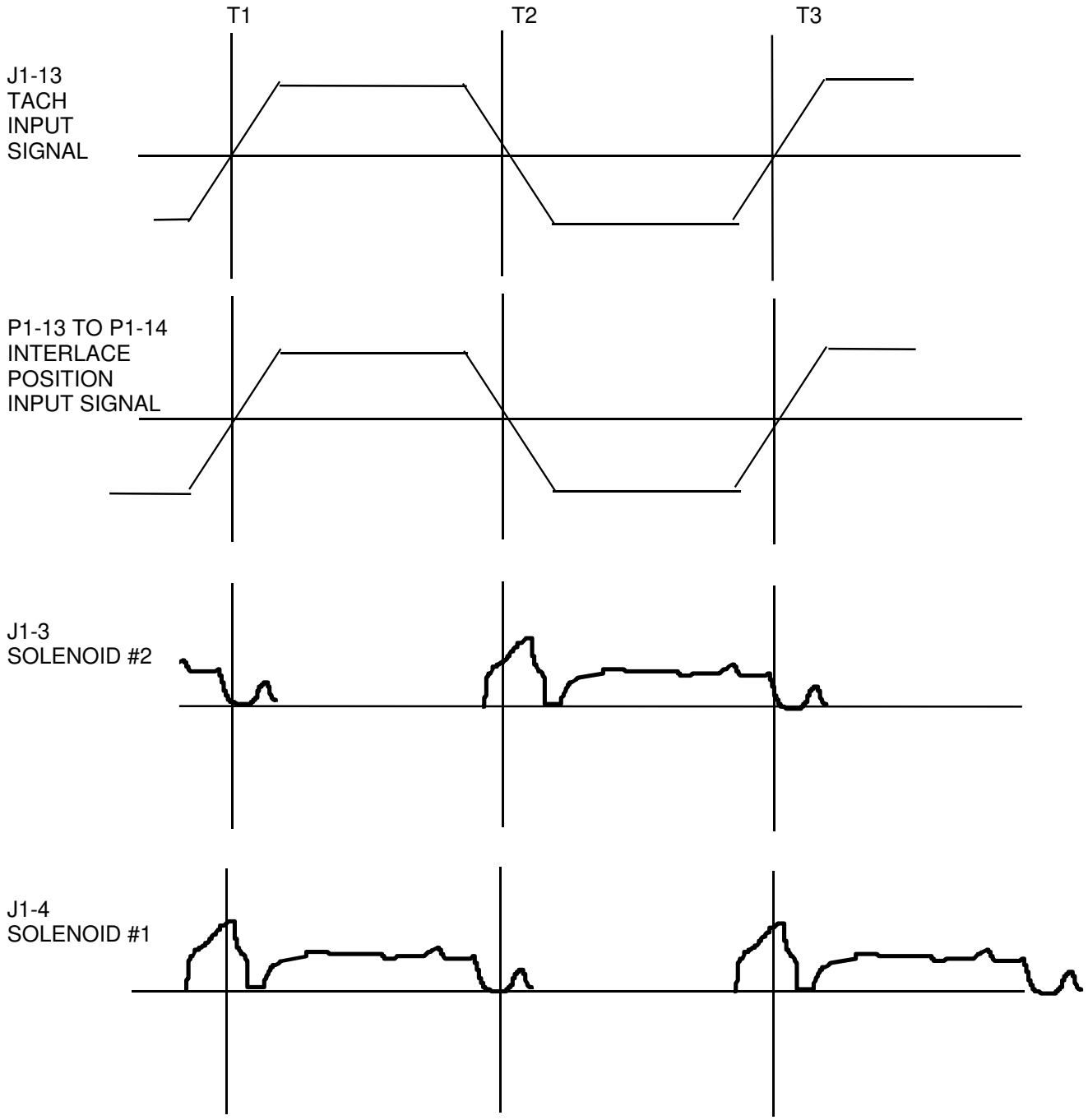
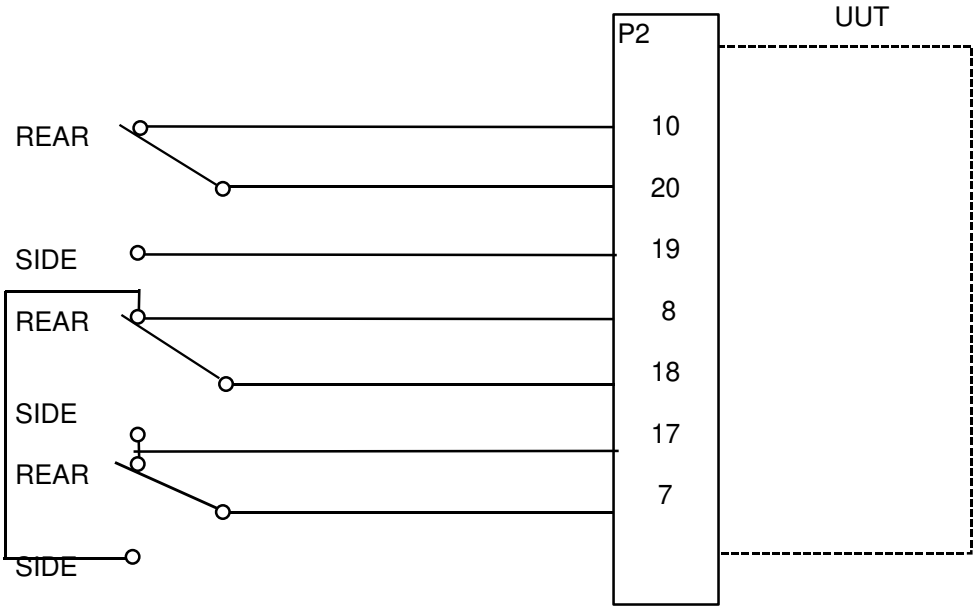


FIGURE 9. Side Phasing Waveforms.

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NOTE: SWITCH SHOWN IN NORMAL (REAR) POSITION.

FIGURE 10. Phasing Logic.

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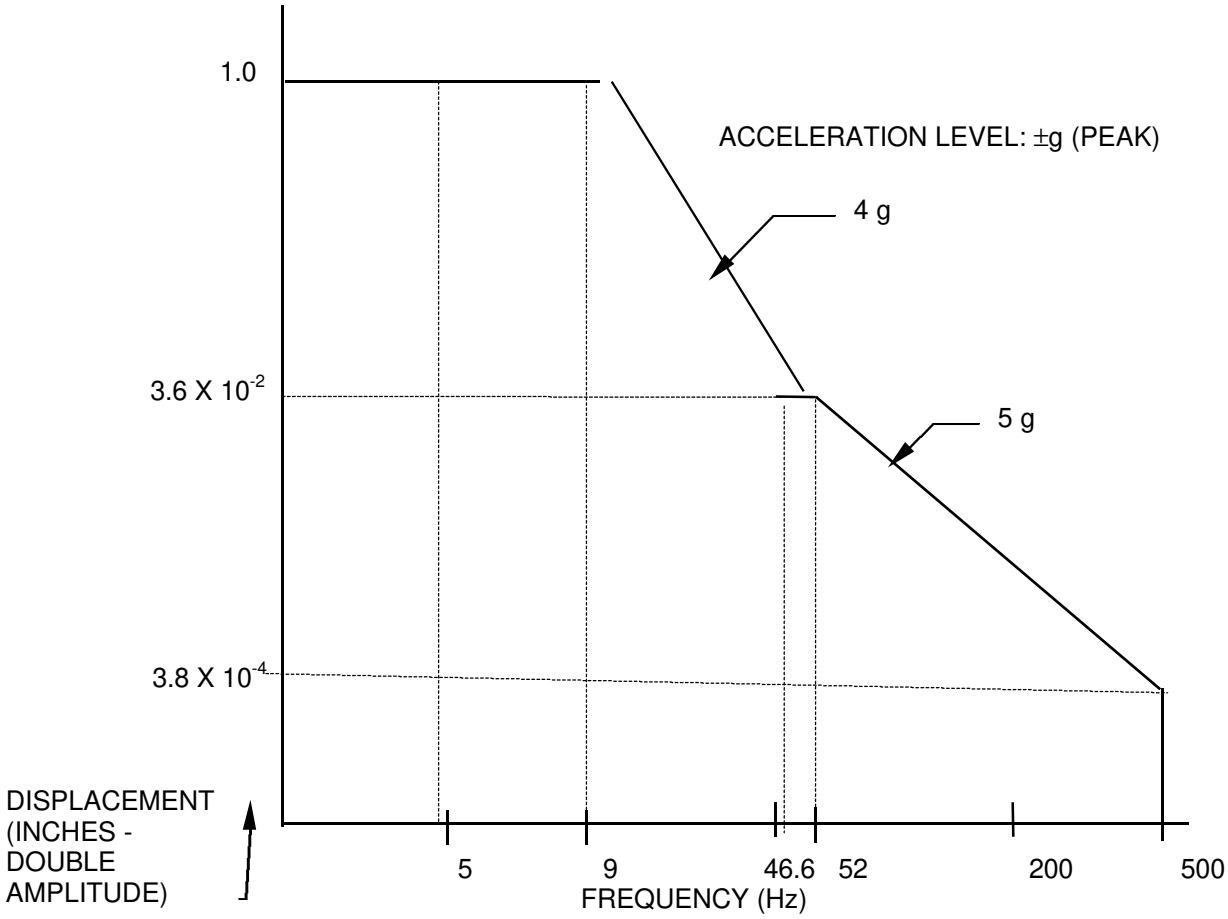


FIGURE 11. Vibration Test Profile.

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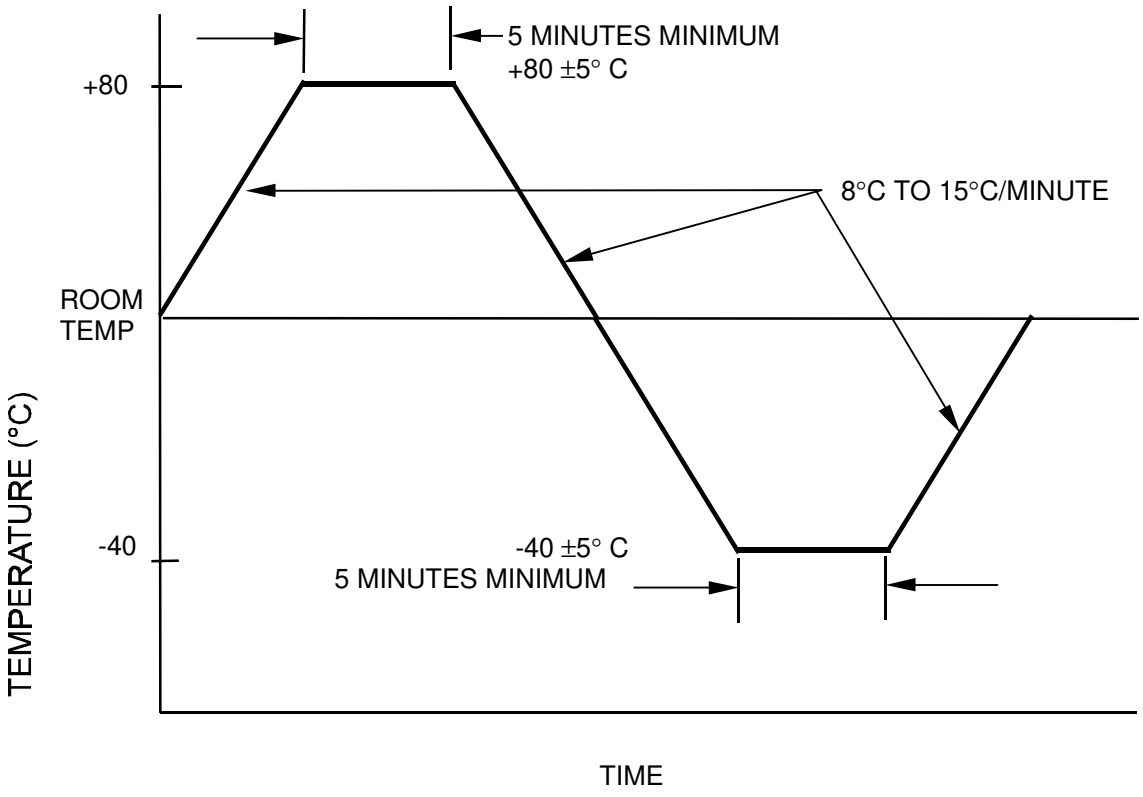


FIGURE 12. One Preconditioning Cycle.

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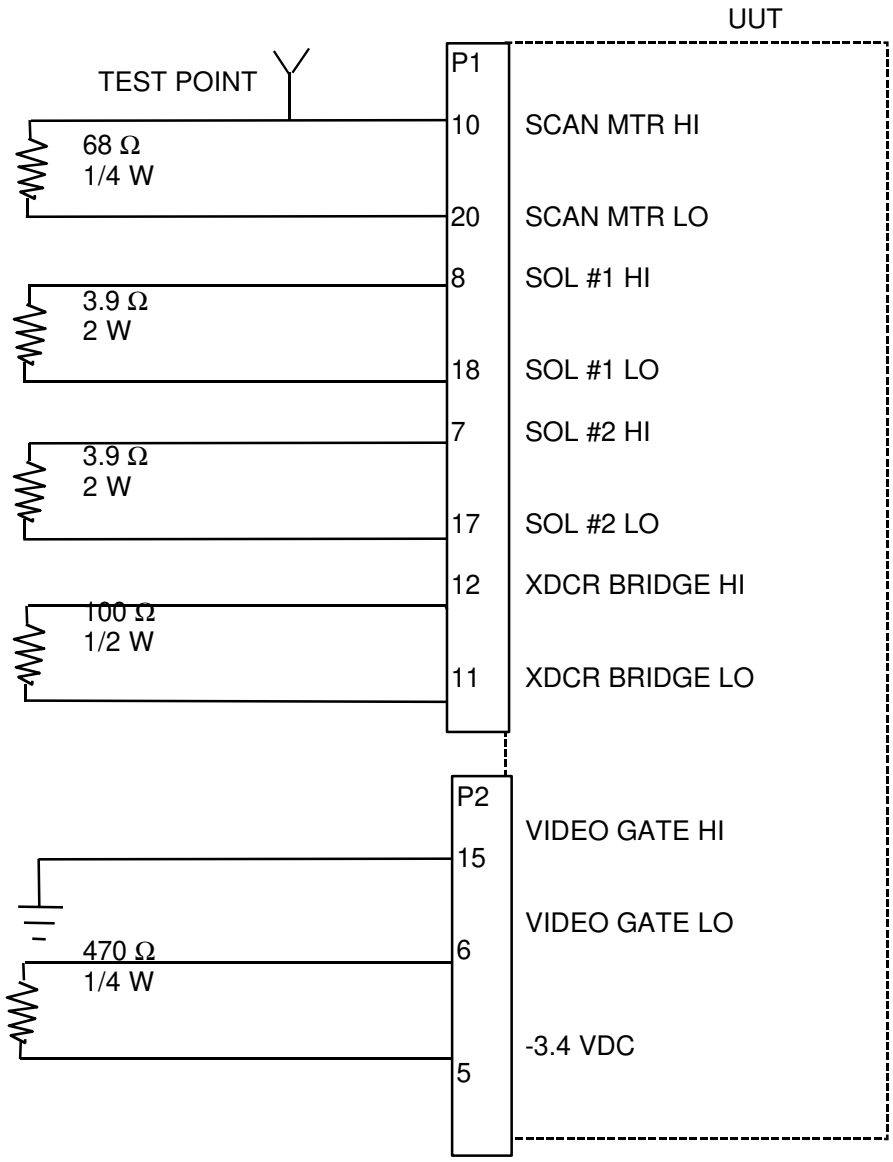
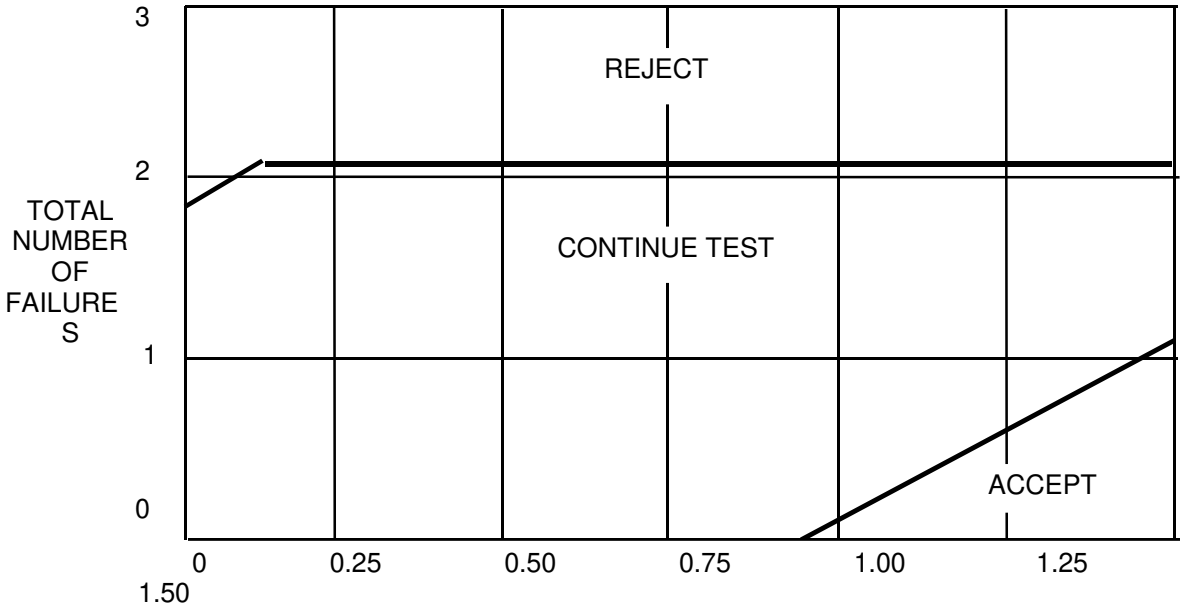


FIGURE 13. Output Loads Preparation.

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TOTAL TEST TIME*

NUMBER OF FAILURES	REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	NA	0.89
1	NA	1.44
2	0.12	1.50
3	1.50	NA

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF EQUIPMENT “ON TIME” AND IS EXPRESSED IN MULTIPLES OF LOWER TEST MTBF θ_1

FIGURE 14. First Article Reliability Accept Reject Criteria.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49168C	**2. DOCUMENT DATE (YYMMDD)** 27 August 1997

3. DOCUMENT TITLE SCAN-INTERLACE, PL-1403/UA, PL-1403A/UA

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9913 (2) AUTOVON 992-9913
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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